Japanese equity megatrends

Growth fields and industries supporting adoption

Field	Supporting areas	Examples of related Japanese companies
AI semiconductors	Production equipment, materials and inspection	Tokyo Electron / Advantest / DISCO / SCREEN Holdings
AI data centers	Electricity supply, power receiving and transformation equipment, cooling	Hitachi / Mitsubishi Electric / Fuji Electric / J-POWER
Optical communications and photonics-electronics convergence	Optical fiber, optical components and R&D	Fujikura / Furukawa Electric / Sumitomo Electric Industries / NTT
AI infrastructure	Data centers and networks	NTT / KDDI / SoftBank / Sakura Internet
Autonomous driving	Sensors, semiconductors and control	Toyota Motor / DENSO / Sony Group / Renesas Electronics

Companies participate differently: product supply, facility operation, R&D and more. A theme connection alone cannot establish the scale of its contribution to sales or profits.

Medium- and long-term technologies

Stages from R&D to adoption vary by technology and use

Field	Supporting areas	Examples of related Japanese companies
Physical AI	Industrial robots and factory automation (FA)	FANUC / Yaskawa Electric / Mitsubishi Electric
Humanoids	Reducers, motors and sensors	Harmonic Drive Systems / Nabtesco / Nidec / Murata Manufacturing
Space	Satellites, rockets and related equipment	Mitsubishi Heavy Industries / IHI / NEC / SKY Perfect JSAT Holdings
Quantum computers	Computing, communications, cryogenics and measurement	Fujitsu / NEC / NTT / Hitachi
Nuclear fusion	Reactor equipment, materials and heavy electrical equipment	Mitsubishi Heavy Industries / IHI / Hitachi / Mitsubishi Electric

Reading the time horizon

Some areas, such as space and industrial robots, are already commercialized; other applications remain centered on R&D. Adoption speed depends on technology, cost, institutions and demand. A single commercialization timeframe cannot easily describe an entire field.

Connections between themes and company analysis

Assess technology, earnings, finances and share prices separately

Industries supporting AI adoption are interconnected

Wider AI use involves semiconductors, data centers, power equipment, cooling and optical communications. Robots and autonomous driving also require motors, sensors, reducers and control technology.

Assessment axis	What to examine
Theme connection	Which products and services are involved? Is there demand shared across several fields?
Earnings contribution	Does demand growth appear in related-business sales, profits and orders?
Finances and cash generation	Check equity ratio, current ratio, operating cash flow and free cash flow.
Share price and corporate value	Assess P/E and P/B alongside growth and profitability.
Forecast uncertainty	Check effects of adoption delays, competition and capital-investment changes on results.

Participation in a growth field is not the same as profit growth or a rising share price. Use the theme map as a starting point, then assess each business and its price.